

Title

Protocol for a Randomized Controlled Trial of Aggressive Beta-Blockade Strategies in Acute Decompensated Cardiac Failure With Rapid Atrial Fibrillation

Abstract

Rapid ventricular response during atrial fibrillation in acute heart failure creates hemodynamic stress and worsens clinical decompensation. This protocol evaluates whether a protocolized approach using a potent beta-blocker combined with additional rate-limiting agents achieves faster heart rate control and greater symptomatic relief compared to conventional rate control strategies. We will enroll approximately 40 hospitalized patients with acute heart failure, systolic dysfunction (ejection fraction <35%), and atrial fibrillation with rapid ventricular response (heart rate >130 beats per minute). Participants are randomly assigned to intensive rate control using beta-blockade plus adjunctive agents or to standard therapy. Endpoints include time to achievement of target heart rate, hemodynamic parameters, dyspnea scores, and adverse events over 24 hours of observation.